

T6C_fuselage_cpp

Fuselage Aerodynamic Coefficients — Physics Formulas by Computation Flow

This document lists every physics formula in the fuselage aerodynamic model, in the exact order they are evaluated: **configuration & derived geometry** → **flow state** → **drag** → **pitching moment** → **yawing moment** → **calibration workflow for when CFD data becomes available**.

Source: F. Nicolosi, P. Della Vecchia, D. Ciliberti, V. Cusati, “Fuselage aerodynamic prediction methods,” *Aerospace Science and Technology*, 55 (2016) 332–343.

Why drag and the two moments use different formula families: Nicolosi’s own method computes C_D from shape factors K_n, K_c, K_t and computes $C_{M0}, C_{M\alpha}, C_{N\beta}$ from an FR -term plus nose/tail corrections – but in the paper **every one of those 12 curves is digitized from the authors’ own STAR-CCM+ CFD sweep** (their Figs. 4–6, 8–16), which this project does not have. The two families differ in how much can be salvaged without it:

- K_n, K_c, K_t : **no** printed numeric value exists anywhere in the paper’s text for these three – chart-only, zero real anchor. Rather than invent one from nothing, drag instead uses the closed-form formulas the paper itself cites as its semi-empirical comparison baseline (Roskam/Kroo, its Eq. 2,6,7,9) – real, validated, and requiring *no* fabricated constant at all (Section 3).
- $C_{M0}, C_{M\alpha}, C_{N\beta}$: each **does** have one real anchor point – the paper’s reference-fuselage worked example is printed as actual numbers, not a chart reading (Eq. 13/14/16). No independently-validated closed-form alternative exists for these (see project README: Roskam’s own fuselage- C_m treatment turns out to be the same Multhopp lineage, and that lineage’s own literature flags it as unreliable without a case-specific correction factor). So each curve here is a line through that one real point, with an **uncalibrated** trend-guessed slope (Sections 5–8).

1 Configuration load (once, at startup)

Raw geometry input (SI units: metres, radians):

$$L_f, d_f, L_n, L_c, L_t, d_b, \theta, \psi, S_w, \bar{c}, b \quad (1)$$

($\bar{c}$ = wing mean aerodynamic chord, b = wingspan – used only for the optional S_w -referenced rescale in Sections 4, 7, 9).

Derived fineness ratios:

$$FR = \frac{L_f}{d_f}, \quad FR_n = \frac{L_n}{d_f}, \quad FR_t = \frac{L_t}{d_f} \quad (2)$$

Frontal area and wetted area (exact nose-cone / cabin-cylinder / tailcone-frustum decomposition – closed-form solid-of-revolution geometry, not an empirical fit):

$$S_{\text{front}} = \frac{\pi}{4} d_f^2 \quad (3)$$

$$S_{\text{wet,nose}} = \pi \frac{d_f}{2} \ell_n, \quad \ell_n = \sqrt{L_n^2 + \left(\frac{d_f}{2}\right)^2} \quad (4)$$

$$S_{\text{wet,cabin}} = \pi d_f L_c \quad (5)$$

$$S_{\text{wet,tail}} = \pi \left(\frac{d_f}{2} + \frac{d_b}{2}\right) \ell_t, \quad \ell_t = \sqrt{L_t^2 + \left(\frac{d_f - d_b}{2}\right)^2} \quad (6)$$

$$S_{\text{wet}} = S_{\text{wet,nose}} + S_{\text{wet,cabin}} + S_{\text{wet,tail}} \quad (7)$$

$FR, FR_n, FR_t, S_{\text{front}}, S_{\text{wet}}, \bullet$ depend only on fixed geometry, so they are computed once in the constructor and cached – `Update()` never recomputes them.

2 Flow state (every call)

$$q = \frac{1}{2}\rho V^2, \quad Re = \frac{\rho V L_f}{\mu} \quad (8)$$

$$C_{Dfp} = \frac{0.455}{(\log_{10} Re)^{2.58} (1 + 0.144 M^2)^{0.65}} \quad (\text{Nicolosi Eq. 2}) \quad (9)$$

3 Fuselage drag — Roskam/Kroo closed-form

Skin friction (Eq. 6; the extracted-PDF text read the fineness-ratio coefficient as 0.0225, but that produces C_{Dsf} alone exceeding the paper’s own reference-case DATCOM total – 0.0025 (matching the well-known Raymer/Hoerner fuselage form factor $(l/d)/400$) reproduces the paper’s Table 3 reference value to within 2.4%, see `main.cpp`’s regression test):

$$C_{Dsf} = C_{Dfp} \left[1 + \frac{60}{(L_f/d_f)^3} + 0.0025 \frac{L_f}{d_f} \right] \frac{S_{\text{wet}}}{S_{\text{front}}} \quad (10)$$

Base drag (Eq. 7, zero if $d_b = 0$):

$$C_{Dbase} = \frac{0.029(d_b/d_f)^3}{\sqrt{C_{Dsf} S_{\text{front}}/S_{\text{wet}}}} \frac{S_{\text{wet}}}{S_{\text{front}}} \quad (11)$$

Upsweep (Eq. 9, Kroo; the extracted-PDF text read the area ratio as S_w/S_{front} , but that produces $C_{Dupsweep}$ alone over double the paper’s own reference-case DATCOM total – inverting it to S_{front}/S_w reproduces Table 3 to within 2.4% together with the corrected C_{Dsf} above; linearized-centreline approximation for $(h/l)_{0.75lt}$, this project’s own addition – Kroo’s formula needs the tailcone centreline’s vertical rise at 75% of tailcone length, non-dimensionalized by tailcone length; approximating the rise as linear from cabin to tail tip makes the ratio independent of L_t):

$$\left(\frac{h}{l} \right)_{0.75lt} \approx 0.75 \tan \theta, \quad C_{Dupsweep} = 0.075 \frac{S_{\text{front}}}{S_w} (0.75 \tan \theta) \quad (12)$$

Windshield contribution (Eq. 8) omitted – needs a graphical Roskam chart, no closed form; a calibration hook is added on top instead:

$$C_{D,\text{fus}}^{(S_{\text{front}})} = C_{Dsf} + C_{Dbase} + C_{Dupsweep} + \text{windshield_drag_increment} \quad (\text{default } 0) \quad (13)$$

4 Drag assembly

Rescaled to the wing-area reference used by `CDo/CDgear` in `t6texan2.xml`:

$$C_{D,\text{fus}}^{(S_w)} = C_{D,\text{fus}}^{(S_{\text{front}})} \frac{S_{\text{front}}}{S_w}, \quad D_{\text{fus}} = q S_{\text{front}} C_{D,\text{fus}}^{(S_{\text{front}})} \quad (14)$$

5 Zero-angle-of-attack pitching moment C_{M0}

Structure (Nicolosi Eq. 11):

$$C_{M0,\text{fus}} = C_{M0,FR}(FR) + \Delta C_{M0,\text{nose}}(FR_n, \psi) + \Delta C_{M0,\text{tail}}(FR_t, \theta) \quad (15)$$

Each term is a line through the **one real anchor point the paper prints as numbers** (Eq. 13, reference fuselage $FR = 8.7$, $FR_n = 1.6$, $\psi = 40^\circ$, $FR_t = 2.8$, $\theta = 14^\circ$), not a chart reading:

$$C_{M0,FR}(FR) = \underbrace{-0.3159}_{\text{Eq.13 anchor}} + s_{FR}(FR - 8.7) \quad (16)$$

$$\Delta C_{M0,nose}(FR_n, \psi) = \underbrace{0.0013}_{\text{Eq.13 anchor}} + s_{n,\psi}(\psi - 40^\circ) + s_{n,FR_n}(FR_n - 1.6) \quad (17)$$

$$\Delta C_{M0,tail}(FR_t, \theta) = \underbrace{0.0013}_{\text{Eq.13 anchor}} + s_{t,\theta}(\theta - 14^\circ) + s_{t,FR_t}(FR_t - 2.8) \quad (18)$$

Slopes s_{FR} , $s_{n,\psi}$, s_{n,FR_n} , $s_{t,\theta}$, s_{t,FR_t} are `TuningParams::pitch` fields, defaulted small (paper: nose correction $\sim 15\%$ of $C_{M0,FR}$, tail correction $< 10\%$, Sec. 5.1) and **uncalibrated**.

6 Pitching moment derivative $C_{M\alpha}$

Same structure (Nicolosi Eq. 12), anchored at Eq. 14:

$$C_{M\alpha,FR}(FR) = \underbrace{0.1815}_{\text{Eq.14 anchor}} + s_{\alpha,FR}(FR - 8.7) \quad (19)$$

$$\Delta C_{M\alpha,nose}(FR_n, \psi) = \underbrace{0.0005}_{\text{Eq.14 anchor}} + \dots \quad (20)$$

$$\Delta C_{M\alpha,tail}(FR_t, \theta) = \underbrace{0.0006}_{\text{Eq.14 anchor}} + \dots \quad (21)$$

(paper: these two corrections change “about 1% or less” with moment reference point, Sec. 5.2 – default slopes accordingly small).

$$C_{M\alpha,fus} = C_{M\alpha,FR} + \Delta C_{M\alpha,nose} + \Delta C_{M\alpha,tail} \quad (22)$$

7 Pitching moment assembly

$$C_{M,fus} = C_{M0,fus} + C_{M\alpha,fus} \alpha \quad (\text{Nicolosi Eq. 10}) \quad (23)$$

Reference is S_{front} and d_f (paper’s own convention):

$$M_{\text{fus}} = q S_{\text{front}} d_f C_{M,fus} \quad (24)$$

Optional rescale to the wing-area/ $\bar{c}$ reference (paper’s Tables 6–7 form):

$$C_{M,fus}^{(S_w, \bar{c})} = C_{M,fus} \frac{S_{\text{front}} d_f}{S_w \bar{c}} \quad (25)$$

8 Yawing moment derivative $C_{N\beta}$

Structure (Nicolosi Eq. 15), anchored at Eq. 16 – note only FR_n/FR_t appear (no ψ, θ dependence stated for yaw, Sec. 6):

$$C_{N\beta,FR}(FR) = \underbrace{0.1817}_{\text{Eq.16 anchor}} + s_{\beta,FR}(FR - 8.7) \quad (26)$$

$$\Delta C_{N\beta,nose}(FR_n) = \underbrace{-0.0003}_{\text{Eq.16 anchor}} + s_{\beta n}(FR_n - 1.6) \quad (27)$$

$$\Delta C_{N\beta,tail}(FR_t) = \underbrace{-0.0002}_{\text{Eq.16 anchor}} + s_{\beta t}(FR_t - 2.8) \quad (28)$$

$$C_{N\beta,fus} = C_{N\beta,FR} + \Delta C_{N\beta,nose} + \Delta C_{N\beta,tail} \quad (29)$$

9 Yawing moment assembly

$$C_{N,\text{fus}} = C_{N\beta,\text{fus}} \beta, \quad N_{\text{fus}} = q S_{\text{front}} d_f C_{N,\text{fus}} \quad (30)$$

Optional rescale to the wing-area/span reference, and the scaling the paper itself specifies if this feeds a whole-aircraft $C_{N\beta}$ (Sec. 6, before adding the vertical tail's contribution):

$$C_{N,\text{fus}}^{(S_w,b)} = C_{N,\text{fus}} \frac{S_{\text{front}} d_f}{S_w b} \quad (31)$$

10 Direction, deliberately left to the caller

D_{fus} is a scalar magnitude opposing the relative wind; $M_{\text{fus}}, N_{\text{fus}}$ are scalar moments about the paper's reference point ($x/L_f=0.50, z/d_f=0.50$). None of the three are resolved into body axes here – the caller rotates/relocates them the same way it treats any other aerodynamic contribution.

11 Calibration workflow — filling in real CFD data later

C_D (Section 3) needs no calibration – Roskam/Kroo is closed-form already. Only $C_{M0}, C_{M\alpha}, C_{N\beta}$ (Sections 5–8) carry uncalibrated slopes. When a real CFD run (STAR-CCM+, OpenFOAM, whatever) on an actual T-6C fuselage geometry becomes available, replace them with fitted ones – **never edit the formulas in `t6c_fuselage.cpp`, only the numbers in `t6c_fuselage_params.yaml`**:

1. Run at least two geometries through CFD (e.g. two different FR at fixed nose/tail, then two different FR_n at fixed FR/FR_t , etc.) and extract $C_{M0}, C_{M\alpha}, C_{N\beta}$ for the whole fuselage at each, the same way Nicolosi generated Figs. 8–16 from many CFD cases.
2. Fit a real slope s_{FR} (etc.) through those points instead of the trend-guess default – with only 2 points this is just $(y_2 - y_1)/(x_2 - x_1)$; with more, a least-squares line.
3. With ≥ 3 –4 real data points per curve, consider replacing the linear trend-guess with the generic `ChartTable1D/ChartTable2D` piecewise-linear interpolation described as future work in the project README, instead of the fixed linear form used here.
4. Update `t6c_fuselage_params.yaml` only; re-run `main.cpp`'s regression test to confirm the new numbers still reproduce whatever CFD points were used to fit them.

Call graph, one call per evaluation (no persistent state):

`FuselageModel::FuselageModel(geometry, tuning)` — Section 1, once, at construction

`FuselageModel::Update(state)` — Sections 2–9, every call, pure function of state (which carries α, β in addition to V, ρ, μ, M) and the cached Section 1 geometry